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GMBA 5885 W04
Strategic Mgt Global Info Sys

Ethical and Social Issues in Information Systems

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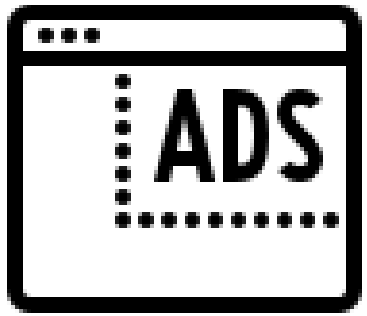
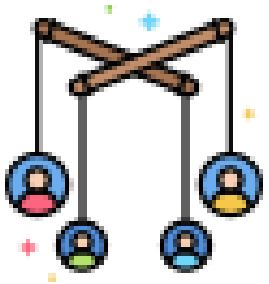
Personalization: Helpful or Harmful?

- When does personalization genuinely improve your life?
  - Share an example that felt useful or time-saving
  - Why did it feel helpful rather than intrusive?
- When does personalization make you uncomfortable?
  - Share a moment when recommendations felt “too accurate” or “too intimate”
  - What triggered that discomfort?
    - Lack of consent?
    - Prediction of sensitive traits?
    - Feeling monitored?



Personalization: Helpful or Harmful?

- Where is your boundary?
  - What data are you comfortable sharing?
  - What data should be off-limits?
  - When must consent be explicit?



- Think deeper: What are the unseen impacts of personalization?
  - Can algorithms nudge your behavior without your awareness?
  - Could repeated recommendations shape your mindset over time?
  - Does personalized content create filter bubbles / echo chambers, limiting what you see?
  - How do these subtle influences affect autonomy and judgment?

Learning Objective

- Discuss IS-related ethical, social, and political (ESP) issues
  - Information rights and privacy
  - Property rights and obligations
  - System quality
  - Accountability and control
  - Quality of life
- Describe principles for ethical conduct

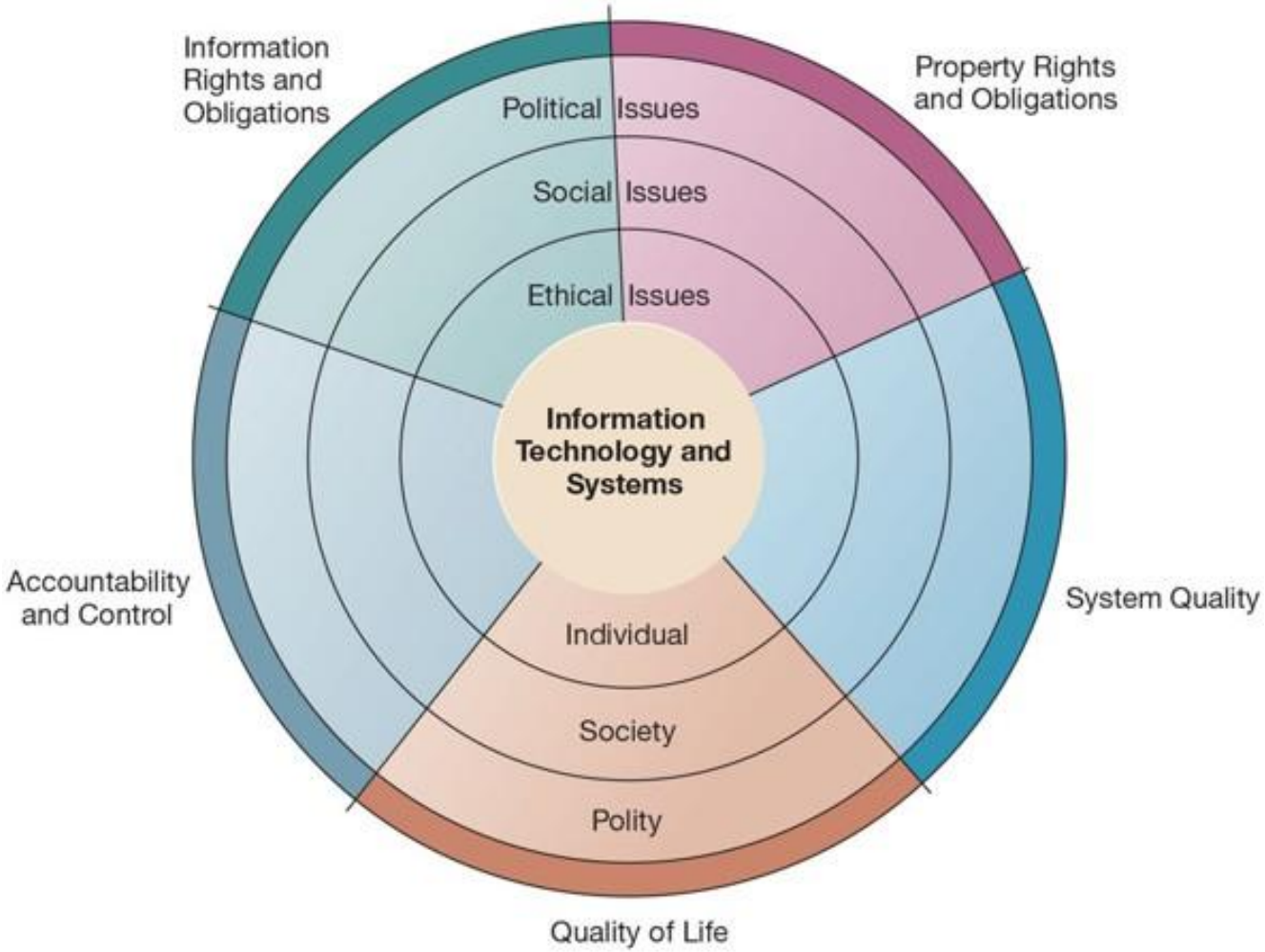
What Do We Mean by “Ethics”

- Ethics
  - Principles of right and wrong that guide individuals and organizational behavior
  - Concerned with
    - What people ought to do
    - How to decide among competing values (ethical dilemma)
    - How to act as a responsible moral agent
- Ethics in business
  - Businesses face recurring dilemmas involving honesty, fairness, transparency, responsibility, and harm avoidance.
  - Ethical failures occur across all industries: fraud, concealment, misuse of information, manipulation.

Why ESP Issues Arise in IS

- IS reshape “how the world works”
  - Power: who controls data and decisions
  - Opportunities: who benefits from digital transformation
  - Rights & obligations: who receives privacy, access, or protection
  - Social values: what count as fair, transparent, or acceptable
- Internet + AI amplifies these tensions
  - Collect massive, granular personal data, often invisibly
  - Operate at speed and scale beyond human oversight
  - Algorithmically influence behavior, choices, and attention
  - Enable both progress (efficiency, innovation) and harm (surveillance, bias, misinformation)

A Model for IS-related ESP Issues



Information Rights and Privacy

- Information systems collect, process, store and distribute data.
- This leads to a fundamental question:

Who owns the data — the individual or the company?

- Two Perspectives
  - Companies: Data is generated on their platforms and systems.
  - Individuals: Data reflects their identity, behavior, and choices.

Information Rights and Privacy

- Think & discuss
  - When you use Amazon, WeChat, TikTok, or airline apps, who do you think owns your data?
  - Should ownership differ by data type: location, purchases, biometrics, browsing history?
  - If you “own” the data, what responsibilities follow?
  - If a company “owns” the data, what obligations should it have?
- Why this matters
  - This debate sits at the core of information rights and privacy.
  - It shapes what individuals can expect, what organizations may do, and how society should regulate digital behavior.

Information Rights and Privacy

- Definition
  - Information rights = the rights that individuals and organizations have over data about themselves
  - Privacy = the claim of individuals to be left alone, free from unwanted surveillance or interference, to control information about themselves

Privacy as a right? A commodity? A state? Control?

- Key question
  - What information rights do individuals and organizations have, and how should they be protected?

Why Privacy Hard to Protect

- Easier, deeper, and broader data collection
  - IS capture data once impossible to observe (e.g., location, behaviors, biometrics)
  - Large-scale, always-on tracking makes surveillance cheap and continuous
  - Interfaces can nudge or mislead users into revealing more
- Data becomes more valuable, exploited
  - Combining and mining datasets generate powerful insights and sensitive inferences
  - Attention, behavior, and predictions become monetizable assets
  - Strong incentives push firms to collect more data, more often, from more places.

Why Privacy Hard to Protect

“很多时候中国人愿意用隐私交换便利性”
------李彦宏

- Privacy paradox
  - Individuals say they care about privacy but often ignore warnings, accept tracking, and trade data for convenience
  - Benefits are immediate; risks are abstract and long-term
  - Privacy choices are confusing, hidden, or time-consuming
  - Most individuals underestimate the potential risk of data disclosure
- Regulation struggles to keep up
  - Technology evolves faster than law
  - Data flows across borders, platforms, and devices
  - Companies exploit vague rules, while users rarely understand or defend their rights

Information Rights and Privacy

Individual privacy rights empowered by the
European General Data Protection Regulation (GDPR)

- Right to information or notice
- Right to access
- Right to data portability
- Right to rectification or correction
- Right to erasure or deletion
- Right to be forgotten
- Right to objection and restriction (or opt out)
- Right to not be subject to automated decisions

The limitations of privacy rights (Solove 2022)

is-ought problem

实然 vs. 应然

Multidimensional Privacy

- Information Systems (IS)
  - Focus: How individuals and organizations manage privacy concerns in digital environments
  - Topics: Privacy calculus, trust, user consent, interface design, privacy-by-design
  - Methods: Surveys, experiments, behavioral studies, system design
- Computer Science / Engineering
  - Focus: Technical mechanisms to protect or exploit data
  - Topics: Encryption, anonymization, differential privacy, PETs, secure architectures
  - Methods: Algorithms, system development, security protocols
- Law & Public Policy
  - Focus: Rights, regulation, enforcement, and societal protection
  - Topics: GDPR, data ownership, liability, consent standards, cross-border data governance
  - Methods: Legal analysis, regulatory frameworks, case law

Multidimensional Privacy

- Economics
  - Focus: Incentives, markets, and trade-offs
  - Topics: Information economic, price discrimination, data valuation, externalities
  - Methods: Modeling, empirical analysis, policy evaluation
- Psychology / Behavioral Science
  - Focus: How people perceive and react to privacy risks
  - Topics: Privacy paradox, cognitive biases, perceived control, trust, risk perception
  - Methods: Experiments, surveys, behavioral observation
- Sociology / Ethics
  - Focus: Social norms, power structures, and moral questions
  - Topics: Surveillance society, digital inequality, autonomy, fairness, collective privacy
  - Methods: Qualitative research, ethical frameworks, cultural analysis

Intellectual Property

Four Key Forms of IP

1. Copyright — Protects Expression
   - Covers creative works (text, music, art, software code)
   - Protects how an idea is expressed, not the idea itself
2. Patents — Protect Ideas & Inventions
   - 20-year monopoly on novel, non-obvious, useful inventions
   - Can apply to software, algorithms, and processes
   - Strong protection, but slow and difficult to obtain

Intellectual Property

Four Key Forms of IP

3. Trademarks — Protect Brands
   - Names, logos, symbols identifying a product or company
   - Prevents confusion and protects brand value (e.g., Google logo)
4. Trade Secrets — Protect “Secret Sauce”
   - Formulas, algorithms, processes kept confidential
   - Protection lasts as long as secrecy is maintained
   - Requires active safeguards (NDAs, restricted access)

IP in Age of AI/GenAI

IS + AI make it effortless to copy, remix, and regenerate content, making uniqueness and ownership harder to define.

- IP in AI Outputs
  - Should AI-generated text, images, or code be protected by IP at all?
  - If so, who owns the output?
    - The user who wrote the prompt?
    - The AI provider who built the model?
    - Or no one (public domain)?
- IP in AI Inputs
  - AI models train on web-scraped images, articles, music, and code, often without permission.
  - Should creators be compensated or allowed to opt out of AI training datasets?
  - If AI can endlessly reproduce an artist's style, does it erode their economic rights?

System Quality

- No system does is perfect
  - How reliable must critical systems be?
  - When is software “good enough” to release?
  - What level of failure is acceptable for society?
- Real-world reliability risks
  - 2023 outages at AWS, Microsoft 365, Oracle OCI disrupted news outlets, airlines, financial systems, and public services.
  - Rare event, but centralized cloud dependence magnifies impact.
- Should firms tolerate these risks? Should providers be more accountable?

System Quality

- Why systems fail
  - Software bugs & design errors
  - Hardware or facility failures
  - Poor input data quality (the most common, often ignored)

  → Zero defects are impossible, complex software always contains bugs.
- Ethical questions
  - Should companies release software knowing it still has bugs?
  - If a system failure harms people, who is responsible?
  - How far should producers go to fix foreseeable problems, especially if perfect reliability is economically impossible?

Accountability and Control

- Key question
  - When an information system causes harm, who should be held accountable?
- Why this matters
  - IS shape decisions, automate processes, and control physical machines.
  - Failures can cause financial loss, safety risks, discrimination, or misinformation.
  - Accountability defines who pays, who fixes, and who prevents future harm.
- Core issues
  - What laws govern misuse of IS?
  - Should system designers, developers, or organizations be liable for harmful outcomes?
  - How do we assign responsibility when multiple stakeholders contribute to a system?

Accountability and Control

- Software liability
  - Poorly designed or unpatched software can cause injury or economic loss.
  - Unlike physical products, software is often not treated as a product → strict liability usually does not apply.
  - Vendors use license agreements to limit liability, and courts often uphold them.
  - Exception: when software is part of a physical product (cars, planes, robots), liability is more likely.
    - *Example: Boeing accepted responsibility for the 737 MAX software failure.*
- AI liability
  - AI operates as a “black box” → hard to explain or trace decisions.
  - Who is responsible when AI makes a harmful or biased decision?
  - EU AI Act (2024): treats AI providers as manufacturers → primary liability for harm; lowers burden of proof for consumers.

Quality of Life

- Concentration of power (big tech)
  - A few firms dominate search, social media, cloud, ads, e-commerce, and AI.
  - Market dominance → economic + political influence, barriers to competition, privacy concerns.
  - Global pushback: Antitrust cases (U.S.), EU Digital Markets Act.
- Digital divide & algorithmic bias
  - Unequal access to devices, broadband, and digital skills → unequal opportunity.
  - “Time-wasting gap”: disadvantaged groups use tech differently (entertainment vs learning).
  - AI trained on biased data can amplify discrimination (loans, hiring, policing).

Quality of Life

- Work & employment disruption
  - Automation, AI, and process redesign eliminate some jobs while creating others
  - Short-term hiring freezes in roles likely to be automated (e.g., customer service)
  - Faster competition → “just-in-time” jobs and workplaces, less stability
- Eroding boundaries: work, family, leisure
  - Ubiquitous computing blurs home–work boundaries
  - Constant connectivity increases stress, reduces personal time, and strains relationships
  - Social media use can contribute to cyberbullying, social comparison, and isolation

Quality of Life

- Key question
  - What values should be preserved in an information driven world?
  - Which social institutions need safeguarding?
  - How does IS/IT reshape culture?
- Discussion
  - AI tools (e.g., ChatGPT) transform how students learn. Do they enhance creativity and understanding, or create dependence?
  - Platforms shape identity, lifestyle, and values. Do they enrich connection and expression, or fuel anxiety and polarization?
  - Are we designing technologies that improve well-being, or just maximize efficiency and profit?

Principles for Ethical Conduct in IS

Four Core Concepts

- **Responsibility:** Individuals must answer for the decisions they make.
- **Accountability:** Systems must allow us to trace who acted and who is responsible.
- **Liability:** Legal mechanisms enable harmed parties to seek compensation.
- **Due Process:** Laws must be known, applied fairly, and allow appeals.

These foundations let us judge whether IS use is ethical, legal, and fair.

Ethical Analysis (5 Steps)

1. **Get the facts** — What actually happened? Who did what, when, and how?
2. **Define the dilemma** — What values or rights are in conflict?
3. **Identify stakeholders** — Who is affected and what do they care about?
4. **List options** — What choices are realistically available?
5. **Evaluate consequences** — Short- and long-term outcomes of each option.

Guiding Ethical Principles

- **Golden Rule:** Treat others as you want to be treated.
- **Kant's Imperative:** If everyone did this, would society function?
- **Slippery Slope:** If repeated, does the action lead somewhere unacceptable?
- **Utilitarian Principle:** Choose the option that produces the greatest good.
- **Risk Aversion:** Choose the option with the least harm.
- **No-Free-Lunch Rule:** Assume all creations have owners who expect compensation.

Being Ethical = Competitive Disadvantage?

Is ethics a constraint, or a strategic choice that builds sustainable success?

- Questions managers often ask
  - If we act ethically, do we risk losing to less ethical competitors?
  - Will others take advantage of our integrity?
  - Is ethics something we consider only after profit and growth?
- Think about these
  - Do ethical choices feel like higher cost, slower speed, or missed opportunities?
  - Have you seen cases where doing the right thing meant giving up revenue or convenience?
  - When is it hardest for a company—or for you—to stay ethical?
- Flip the perspective
  - What hidden costs arise when firms cut ethical corners? (trust, reputation, regulation, morale, long-term value)
  - Can ethics become a competitive advantage? In what ways?

Summary

- Why issues arise
  - IS and AI reshape power, rights, and behavior, and operate at scale and speed that amplify risks.
- Key domains
  - Information rights & privacy, intellectual property, system quality & reliability, accountability & control, quality of life impacts
- Ethical decision tools
  - Core concepts: responsibility, accountability, liability, due process
  - Ethical analysis: understand facts, values, stakeholders, options, consequences
- Managerial Insight
  - Ethics is not a constraint — it's part of sustainable, trustworthy IS management.